

Anna Ma

Associate Professor

Department of Mathematics

University of California, Irvine

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Website: anna-math.github.io

My research focuses on the problems that arise in mathematical data science. I am particularly interested in designing and analyzing iterative algorithms for large-scale data, as well as leveraging and developing tools in numerical linear algebra, signal processing, machine learning, and probability.

Academic Positions

Associate Professor

July 2026 - Current

Department of Mathematics
University of California, Irvine

Assistant Professor

July 2022 - June 2026

Department of Mathematics
University of California, Irvine

Visiting Assistant Professor

July 2019 - June 2022

Department of Mathematics
University of California, Irvine
Faculty Mentor: Roman Vershynin

UC Chancellor's Postdoctoral Fellow

July 2018 - June 2019

Department of Mathematics,
University of California, San Diego
Faculty Mentor: Rayan Saab

Education

PhD Claremont Graduate University, Computational Science

Sept. 2013 – June 2018

Jointly with San Diego State University
Dissertation: "Stochastic Iterative Algorithms for Large-Scale Data"
Thesis Adviser: Deanna Needell

BS University of California, Los Angeles, Mathematics

Sept. 2009 – June 2013

Minor in Statistics
Specialization in Computing

Publications

Accepted Publications

- [1] A. Ma, A. Flenner, D. Needell, and A. G. Percus. "Improving image clustering using sparse text and the wisdom of the crowds". *2014 48th Asilomar Conference on Signals, Systems and Computers*. IEEE. 2014, pp. 1555–1557. doi: [10.1109/ACSSC.2014.7094725](https://doi.org/10.1109/ACSSC.2014.7094725).
- [2] A. Ma, D. Needell, and A. Ramdas. "Convergence properties of the randomized extended Gauss–Seidel and Kaczmarz methods". *SIAM Journal on Matrix Analysis and Applications* 36.4 (2015), pp. 1590–1604. doi: [10.1137/15M1014425](https://doi.org/10.1137/15M1014425).
- [3] D. Baron, A. Ma, D. Needell, C. Rush, and T. Woolf. "Conditional approximate message passing with side information". *2017 51st Asilomar Conference on Signals, Systems, and Computers*. IEEE. 2017, pp. 430–434. doi: [10.1109/ACSSC.2017.8335374](https://doi.org/10.1109/ACSSC.2017.8335374).
- [4] A. Ma and D. Needell. "A gradient descent approach for incomplete linear systems". *2018 52nd Asilomar Conference on Signals, Systems, and Computers*. IEEE. 2018, pp. 764–768. doi: [10.1109/ACSSC.2018.8645206](https://doi.org/10.1109/ACSSC.2018.8645206).

- [5] A. Ma, D. Needell, and A. Ramdas. “Iterative methods for solving factorized linear systems”. *SIAM Journal on Matrix Analysis and Applications* 39.1 (2018), pp. 104–122. doi: [10.1137/17M1115678](https://doi.org/10.1137/17M1115678).
- [6] S. Ravishankar, A. Ma, and D. Needell. “Analysis of Fast Alternating Minimization for Structured Dictionary Learning”. *2018 Information Theory and Applications Workshop (ITA)*. IEEE. 2018, pp. 1–9. doi: [10.1109/ITA.2018.8503246](https://doi.org/10.1109/ITA.2018.8503246).
- [7] Y. Van Gennip, B. Hunter, A. Ma, D. Moyer, R. De Vera, and A. L. Bertozzi. “Unsupervised record matching with noisy and incomplete data”. *International Journal of Data Science and Analytics* 6.2 (2018), pp. 109–129. doi: [10.1007/s41060-018-0129-7](https://doi.org/10.1007/s41060-018-0129-7).
- [8] N. Durgin, R. Grotheer, C. Huang, S. Li, A. Ma, D. Needell, and J. Qin. “Compressed Anomaly Detection with Multiple Mixed Observations”. *Research in Data Science*. Springer, 2019, pp. 211–237. doi: [10.1007/978-3-030-11566-1_10](https://doi.org/10.1007/978-3-030-11566-1_10).
- [9] N. Durgin, R. Grotheer, C. Huang, S. Li, A. Ma, D. Needell, and J. Qin. “Fast hyperspectral diffuse optical imaging method with joint sparsity”. *2019 41st Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*. IEEE. 2019, pp. 4758–4761. doi: [10.1109/EMBC.2019.8857069](https://doi.org/10.1109/EMBC.2019.8857069).
- [10] N. Durgin, R. Grotheer, C. Huang, S. Li, A. Ma, D. Needell, and J. Qin. “Jointly Sparse Signal Recovery with Prior Info”. *2019 53rd Asilomar Conference on Signals, Systems, and Computers*. IEEE. 2019, pp. 645–649. doi: [10.1109/IEEECONF59524.2023.10477079](https://doi.org/10.1109/IEEECONF59524.2023.10477079).
- [11] N. Durgin, R. Grotheer, C. Huang, S. Li, A. Ma, D. Needell, and J. Qin. “Sparse randomized Kaczmarz for support recovery of jointly sparse corrupted multiple measurement vectors”. *Research in Data Science*. Springer, 2019, pp. 1–14. doi: [10.1007/978-3-030-11566-1_1](https://doi.org/10.1007/978-3-030-11566-1_1).
- [12] A. Ma, Y. Zhou, C. Rush, D. Baron, and D. Needell. “An approximate message passing framework for side information”. *IEEE Transactions on Signal Processing* 67.7 (2019), pp. 1875–1888. doi: [10.1109/TSP.2019.2899286](https://doi.org/10.1109/TSP.2019.2899286).
- [13] D. Needell and A. Ma. “Stochastic Gradient Descent for Linear Systems with Missing Data”. *Numerical Mathematics: Theory, Methods and Applications* 12.1 (2019). doi: [10.4208/nmtma.0A-2018-0066](https://doi.org/10.4208/nmtma.0A-2018-0066).
- [14] R. Grotheer, S. Li, A. Ma, D. Needell, and J. Qin. “Stochastic iterative hard thresholding for low-Tucker-rank tensor recovery”. *2020 Information Theory and Applications Workshop (ITA)*. IEEE. 2020, pp. 1–5. doi: [10.1109/ITA50056.2020.9244965](https://doi.org/10.1109/ITA50056.2020.9244965).
- [15] S. Ravishankar, A. Ma, and D. Needell. “Analysis of fast structured dictionary learning”. *Information and Inference: A Journal of the IMA* 9.4 (2020), pp. 785–811. doi: [10.1093/imaiai/iaz028](https://doi.org/10.1093/imaiai/iaz028).
- [16] J. A. De Loera, J. Haddock, A. Ma, and D. Needell. “Data-driven algorithm selection and tuning in optimization and signal processing”. *Annals of Mathematics and Artificial Intelligence* 89.7 (2021), pp. 711–735. doi: [10.1007/s10472-020-09717-z](https://doi.org/10.1007/s10472-020-09717-z).
- [17] N. Durgin, R. Grotheer, C. Huang, S. Li, A. Ma, D. Needell, and J. Qin. “A Simple Recovery Framework for Signals with Time-Varying Sparse Support”. *Advances in Data Science*. Springer, 2021, pp. 211–230. doi: [10.1007/978-3-030-79891-8_9](https://doi.org/10.1007/978-3-030-79891-8_9).
- [18] J. Haddock and A. Ma. “Greed works: An improved analysis of sampling Kaczmarz–Motzkin”. *SIAM Journal on Mathematics of Data Science* 3.1 (2021), pp. 342–368. doi: [10.1137/19M1307044](https://doi.org/10.1137/19M1307044).
- [19] A. Ma and D. Molitor. “Randomized Kaczmarz for tensor linear systems”. *BIT Numerical Mathematics* (2021), pp. 1–24. doi: [10.1007/s10543-021-00877-w](https://doi.org/10.1007/s10543-021-00877-w).
- [20] J. Qin, S. Li, D. Needell, A. Ma, R. Grotheer, C. Huang, and N. Durgin. “Stochastic greedy algorithms for multiple measurement vectors”. *Inverse Problems & Imaging* 15.1 (2021), p. 79. doi: [10.3934/ipi.2020066](https://doi.org/10.3934/ipi.2020066).
- [21] R. Grotheer, S. Li, A. Ma, D. Needell, and J. Qin. “Iterative hard thresholding for low cp-rank tensor models”. *Linear and Multilinear Algebra* 70.22 (2022), pp. 7452–7468. doi: [10.1080/03081087.2021.1992335](https://doi.org/10.1080/03081087.2021.1992335).
- [22] E. Lybrand, A. Ma, and R. Saab. “On the number of faces and radii of cells induced by Gaussian spherical tessellations”. *Applied and Computational Harmonic Analysis* 56 (2022), pp. 176–188. doi: [10.1016/j.acha.2021.08.005](https://doi.org/10.1016/j.acha.2021.08.005).
- [23] K. Dover, Z. Cang, A. Ma, Q. Nie, and R. Vershynin. “AVIDA: An alternating method for visualizing and integrating data”. *Journal of Computational Science* 68 (2023), p. 101998. doi: [10.1016/j.jocs.2023.101998](https://doi.org/10.1016/j.jocs.2023.101998).
- [24] R. Grotheer, S. Li, A. Ma, D. Needell, and J. Qin. “Stochastic natural thresholding algorithms”. *2023 57th Asilomar Conference on Signals, Systems, and Computers*. IEEE. 2023, pp. 832–836. doi: [10.1109/IEEECONF59524.2023.10477079](https://doi.org/10.1109/IEEECONF59524.2023.10477079).

- [25] J. Haddock, A. Ma, and E. Rebrova. “On subsampled quantile randomized Kaczmarz”. *2023 59th Annual Allerton Conference on Communication, Control, and Computing (Allerton)*. IEEE. 2023, pp. 1–8. doi: [10.1109/Allerton58177.2023.10313381](https://doi.org/10.1109/Allerton58177.2023.10313381) .
- [26] C. Huynh, A. Ma, and M. Strand. “Block-missing data in linear systems: An unbiased stochastic gradient descent approach”. *2023 59th Annual Allerton Conference on Communication, Control, and Computing (Allerton)*. IEEE. 2023, pp. 1–7. doi: [10.1109/Allerton58177.2023.10313429](https://doi.org/10.1109/Allerton58177.2023.10313429) .
- [27] A. Ma, D. Stöger, and Y. Zhu. “Robust recovery of low-rank matrices and low-tubal-rank tensors from noisy sketches”. *SIAM Journal on Matrix Analysis and Applications* 44.4 (2023), pp. 1566–1588. doi: [10.1137/22M150071X](https://doi.org/10.1137/22M150071X) .
- [28] E. H. Bergou, S. Bouchrouite, A. Dutta, X. Li, and A. Ma. “A Note on the Randomized Kaczmarz Algorithm for Solving Doubly Noisy Linear Systems”. *SIAM Journal on Matrix Analysis and Applications* 45.2 (2024), pp. 992–1006. doi: [10.1137/23M155712X](https://doi.org/10.1137/23M155712X) .
- [29] R. Grotheer, S. Li, A. Ma, D. Needell, and J. Qin. “Iterative singular thresholding algorithms for tensor recovery”. *Inverse Problems and Imaging* 18.4 (2024), pp. 889–907. doi: [10.3934/ipi.2023059](https://doi.org/10.3934/ipi.2023059) .
- [30] E. Battaglia and A. Ma. “Quantile Randomized Kaczmarz with Time-Varying Noise in Signal”. *To appear in Proceedings of the IEEE Asilomar Conference on Signals, Systems and Computers*. 2025.
- [31] E. Battaglia and A. Ma. “Reverse Quantile-RK and Its Application to Quantile-RK”. *Numerical Linear Algebra with Applications* 32.3 (2025), e70024. doi: <https://doi.org/10.1002/nla.70024> .
- [32] H. Luo and A. Ma. “Frontal Slice Approaches for Tensor Linear Systems”. *Numerical Mathematics: Theory, Methods and Applications* 18.2 (2025), pp. 353–394. doi: [10.4208/nmtma.0A-2024-0096](https://doi.org/10.4208/nmtma.0A-2024-0096) .
- [33] A. Ma, D. Needell, and A. Xue. “Stochastic Gradient Descent on Tensors with Missing Data”. *To appear in Proceedings of the AAAI 2025 Workshop on CoLoRAI*. 2025.
- [34] E. Battaglia and A. Ma. “Quantile-RK and double quantile-RK error horizon analysis”. *Linear Algebra and its Applications* 736 (2026), pp. 284–308. doi: [10.1016/j.laa.2026.01.032](https://doi.org/10.1016/j.laa.2026.01.032) .
- [35] J.-F. Cai, J. Chen, A. Ma, and T. Wu. “On the Subsample Size of Quantile-Based Randomized Kaczmarz”. *SIAM Journal on Matrix Analysis and Applications* 47.2 (2026), pp. 802–823. doi: [10.1137/25M1785678](https://doi.org/10.1137/25M1785678) .
- [36] H. Luo, A. Ma, L. Stephan, and Y. Zhu. “Wedge Sampling: Efficient Tensor Completion with Nearly-Linear Sample Complexity”. *Proceedings of Thirty-Ninth Conference on Learning Theory*. Ed. by S. Hanneke and T. Lattimore. Vol. 336. Proceedings of Machine Learning Research. PMLR, 29 Jun–03 Jul 2026, pp. 4883–4884. URL: <https://proceedings.mlr.press/v336/luo26a.html>.

Conference Publications

- [1] A. Ma, A. Flenner, D. Needell, and A. G. Percus. “Improving image clustering using sparse text and the wisdom of the crowds”. *2014 48th Asilomar Conference on Signals, Systems and Computers*. IEEE. 2014, pp. 1555–1557. doi: [10.1109/ACSSC.2014.7094725](https://doi.org/10.1109/ACSSC.2014.7094725) .
- [3] D. Baron, A. Ma, D. Needell, C. Rush, and T. Woolf. “Conditional approximate message passing with side information”. *2017 51st Asilomar Conference on Signals, Systems, and Computers*. IEEE. 2017, pp. 430–434. doi: [10.1109/ACSSC.2017.8335374](https://doi.org/10.1109/ACSSC.2017.8335374) .
- [4] A. Ma and D. Needell. “A gradient descent approach for incomplete linear systems”. *2018 52nd Asilomar Conference on Signals, Systems, and Computers*. IEEE. 2018, pp. 764–768. doi: [10.1109/ACSSC.2018.8645206](https://doi.org/10.1109/ACSSC.2018.8645206) .
- [6] S. Ravishankar, A. Ma, and D. Needell. “Analysis of Fast Alternating Minimization for Structured Dictionary Learning”. *2018 Information Theory and Applications Workshop (ITA)*. IEEE. 2018, pp. 1–9. doi: [10.1109/ITA.2018.8503246](https://doi.org/10.1109/ITA.2018.8503246) .
- [9] N. Durgin, R. Grotheer, C. Huang, S. Li, A. Ma, D. Needell, and J. Qin. “Fast hyperspectral diffuse optical imaging method with joint sparsity”. *2019 41st Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*. IEEE. 2019, pp. 4758–4761. doi: [10.1109/EMBC.2019.8857069](https://doi.org/10.1109/EMBC.2019.8857069) .
- [10] N. Durgin, R. Grotheer, C. Huang, S. Li, A. Ma, D. Needell, and J. Qin. “Jointly Sparse Signal Recovery with Prior Info”. *2019 53rd Asilomar Conference on Signals, Systems, and Computers*. IEEE. 2019, pp. 645–649. doi: [10.1109/IEEECONF44664.2019.9048818](https://doi.org/10.1109/IEEECONF44664.2019.9048818) .
- [14] R. Grotheer, S. Li, A. Ma, D. Needell, and J. Qin. “Stochastic iterative hard thresholding for low-Tucker-rank tensor recovery”. *2020 Information Theory and Applications Workshop (ITA)*. IEEE. 2020, pp. 1–5. doi: [10.1109/ITA50056.2020.9244965](https://doi.org/10.1109/ITA50056.2020.9244965) .

- [24] R. Grotheer, S. Li, A. Ma, D. Needel, and J. Qin. “Stochastic natural thresholding algorithms”. *2023 57th Asilomar Conference on Signals, Systems, and Computers*. IEEE. 2023, pp. 832–836. doi: [10.1109/IEEECONF59524.2023.10477079](https://doi.org/10.1109/IEEECONF59524.2023.10477079) .
- [25] J. Haddock, A. Ma, and E. Rebrova. “On subsampled quantile randomized Kaczmarz”. *2023 59th Annual Allerton Conference on Communication, Control, and Computing (Allerton)*. IEEE. 2023, pp. 1–8. doi: [10.1109/Allerton58177.2023.10313381](https://doi.org/10.1109/Allerton58177.2023.10313381) .
- [26] C. Huynh, A. Ma, and M. Strand. “Block-missing data in linear systems: An unbiased stochastic gradient descent approach”. *2023 59th Annual Allerton Conference on Communication, Control, and Computing (Allerton)*. IEEE. 2023, pp. 1–7. doi: [10.1109/Allerton58177.2023.10313429](https://doi.org/10.1109/Allerton58177.2023.10313429) .
- [30] E. Battaglia and A. Ma. “Quantile Randomized Kaczmarz with Time-Varying Noise in Signal”. *To appear in Proceedings of the IEEE Asilomar Conference on Signals, Systems and Computers*. 2025.
- [33] A. Ma, D. Needell, and A. Xue. “Stochastic Gradient Descent on Tensors with Missing Data”. *To appear in Proceedings of the AAAI 2025 Workshop on CoLoRAL*. 2025.
- [36] H. Luo, A. Ma, L. Stephan, and Y. Zhu. “Wedge Sampling: Efficient Tensor Completion with Nearly-Linear Sample Complexity”. *Proceedings of Thirty-Ninth Conference on Learning Theory*. Ed. by S. Hanneke and T. Lattimore. Vol. 336. Proceedings of Machine Learning Research. PMLR, 29 Jun–03 Jul 2026, pp. 4883–4884. URL: <https://proceedings.mlr.press/v336/luo26a.html>.

Preprints

- [37] Y.-T. Liao, H. Luo, and A. Ma. “Efficient and robust Bayesian selection of hyperparameters in dimension reduction for visualization” (2023). URL: <https://arxiv.org/abs/2306.00357>.
- [38] E. H. Bergou, S. Boucherouite, A. Dutta, X. Li, and A. Ma. “Where Have All the Kaczmarz Iterates Gone?” *arXiv preprint arXiv:2510.08563* (2025). doi: [10.48550/arXiv.2510.08563](https://doi.org/10.48550/arXiv.2510.08563) .
- [39] H. Luo, A. Ma, L. Stephan, and Y. Zhu. “Wedge Sampling: Efficient Tensor Completion with Nearly-Linear Sample Complexity”. *arXiv preprint arXiv:2602.05869* (2026). doi: [10.48550/arXiv.2602.05869](https://doi.org/10.48550/arXiv.2602.05869) .

Other Works

- [40] A. Ma. “Geometry, Intuition, and the Kaczmarz Algorithm”. *Girls’ Angle Bulletin* 14.6 (Aug. 2021). URL: <https://www.girlsangle.org/page/bulletin-archive/GABv14n06E.pdf>.
- [41] A. Ma. “Getting to the Bottom of It: Gradient Descent and its Variants”. *Girls’ Angle Bulletin* 15.1 (Oct. 2021). URL: <https://www.girlsangle.org/page/bulletin-archive/GABv15n01E.pdf>.
- [42] A. Ma. “When data goes missing”. *Girls’ Angle Bulletin* 15.2 (Dec. 2021). URL: <https://www.girlsangle.org/page/bulletin-archive/GABv15n02E.pdf>.
- [43] A. Ma. “What Is Data: Representing Information as a Matrix”. *Girls’ Angle Bulletin* 15.3 (Feb. 2022). URL: <https://www.girlsangle.org/page/bulletin-archive/GABv15n03E.pdf>.
- [44] A. Ma. “Data Collection: Challenges and Workarounds”. *Girls’ Angle Bulletin* 15.4 (Apr. 2022). URL: <https://www.girlsangle.org/page/bulletin-archive/GABv15n04E.pdf>.
- [45] A. Ma. “Data: Classical and Modern Data Visualization Methods”. *Girls’ Angle Bulletin* 15.5 (June 2022). URL: <https://www.girlsangle.org/page/bulletin-archive/GABv15n05E.pdf>.
- [46] A. Ma. “Supervised Machine Learning: Classification with Candy”. *Girls’ Angle Bulletin* 15.6 (Aug. 2022). URL: <https://www.girlsangle.org/page/bulletin-archive/GABv15n06E.pdf>.
- [47] A. Ma. “The Perceptron Algorithm The Perceptron Algorithm”. *Girls’ Angle Bulletin* 16.1 (Oct. 2022). URL: <https://www.girlsangle.org/page/bulletin-archive/GABv16n01E.pdf>.
- [48] A. Ma. “Natural Language Processing Natural Language Processing”. *Girls’ Angle Bulletin* 16.2 (Dec. 2022). URL: <https://www.girlsangle.org/page/bulletin-archive/GABv16n02E.pdf>.
- [49] A. Ma. “Natural Language Processing, Part 2: Language Models”. *Girls’ Angle Bulletin* 16.4 (Apr. 2023). URL: <https://www.girlsangle.org/page/bulletin-archive/GABv16n04E.pdf>.
- [50] A. Ma. “Natural Language Processing, Part 3: From Language Models to AI Chatbots”. *Girls’ Angle Bulletin* 16.5 (June 2023). URL: <https://www.girlsangle.org/page/bulletin-archive/GABv16n05E.pdf>.

Professional Activities

Fairness and foundations in machine learning

American Institute of Mathematics
Co-organizer

July 2026
Pasadena, CA

Semester Program: Stochastic and Randomized Algorithms in Scientific Computing: Foundations and Applications

Institute for Computational and Experimental Research in Mathematics (ICERM)
Invited participant for a semester program

Spring 2026
Providence, RI

Research Collaboration Workshop on Randomized Numerical Linear Algebra

Institute for Pure and Applied Mathematics (IPAM)
Invited participant for week-long working group
Topic: "Randomization in transformer models"

August 2025
Los Angeles, CA

Research Collaboration Workshop in the Science of Data and Mathematics

University of North Carolina at Chapel Hill
Invited participant for week-long working group
Topic: "Randomized Methods for Tensors"

August 2025
Chapel Hill, NC

Collaborate@ICERM

Institute for Computational and Experimental Research in Mathematics (ICERM)
Funded proposal for week-long working group
Topic: "Noise is not Always Bad: A Case Study with the Kaczmarz Algorithm"

July 2025
Providence, RI

Summer Collaborations Program

Institute for Advanced Study (IAS)
Funded proposal for two-week-long working group
Topic: "Partial Information Methods for Large-scale Linear Systems"

July 2025
Princeton, NJ

FOCUS Writing Workshop

University of California, Berkeley
Invited to participate in a two-day workshop with grant writing consultant

April 2025
Berkeley, CA

Mathematics of Intelligences

Institute for Pure and Applied Mathematics (IPAM)
Invited participant for week-long working group
Topic: "Analyzing High-dimensional Traces of Intelligent Behavior"

September 2024
Los Angeles, CA

Research Collaboration Workshop: Women in Data Science and Mathematics

Institute for Pure and Applied Mathematics (IPAM)
Invited participant for week-long working group
Topic: "Graph-based Active Learning"

August 2023
Los Angeles, CA

Women and Mathematics Program

Institute for Advanced Study (IAS)
Invited participant for week-long working group
Topic: "The Mathematics of Machine Learning"

May 2022
Los Angeles, CA

Summer Research in Mathematics

Mathematical Sciences Research Institute (MSRI)
Funded proposal for week-long working group
Topic: "Practical Problems in Distributed Sensing"

June 2019
Berkeley, CA

Women in Data Science and Mathematics

Institute for Computational and Experimental Research in Mathematics (ICERM)
Invited participant for week-long working group
Topic: "Tensor Tools for Multiway Data Analysis"

June 2019
Providence, RI

Visiting Graduate Researcher

University of California, Los Angeles
Spent one quarter at UCLA as a visiting graduate researcher

Spring 2018
Los Angeles, CA

Topic: "Sparse and Stochastic Optimization"

Collaborate@ICERM

Institute for Computational and Experimental Research in Mathematics (ICERM)

Funded proposal for week-long working group

Topic: "Control and Analysis of Large-Scale Time-Varying Data"

July 2018
Providence, RI

Semester Program: Geometric Functional Analysis and Applications

Mathematical Sciences Research Institute (MSRI)

Invited participant for a semester program

Fall 2017
Berkeley, CA

Women in Data Science and Mathematics

Institute for Computational and Experimental Research in Mathematics (ICERM)

Invited participant for week-long working group

Topic: "Stochastic Signal Processing for High Dimensional Data"

July 2017
Providence, RI

Gene Golab SIAM Summer School

Akademie Berlin-Schmöckwitz, Germany

Invited participant for two-week summer school

Topic: "Sparse Recovery and Optimization"

July 2017
Berlin, Germany

Bayes Impact

Data Science Fellow for 501(c)(3) non-profit Start-up

Worked with an Education Technology company to predict standardized test scores

Spring 2015
San Francisco, CA

Graduate Research in Industry Projects (GRIPS)

Institute for Pure and Applied Mathematics (IPAM)

Graduate researcher for inaugural GRIPS program

Topic: Gas Networks with Open Grid Europe

Summer 2014
Berlin, Germany

Research in Industry Projects (RIPS)

Institute for Pure and Applied Mathematics (IPAM)

Undergraduate researcher and project manager for RIPS program

Topic: Linking Social Media to Crime and Disorder

Summer 2013
Los Angeles, CA

Research Experience for Undergraduate Students

University of California, Los Angeles

Undergraduate researcher

Topic: Social Networks and Duplicate Detection

Summer 2012
Los Angeles, CA

Grants, Honors, and Awards

NSF DMS-2606335

Collaborative Research: Scalable Methods for Incomplete Tensors: To Complete or Not to Complete?

Total award estimate: \$91,400

09/2026-08/2039

Travel Support for Mathematicians

Simons Foundation travel support for Mathematicians

Total award estimate: \$42,000

09/2025-08/2030

Rose Hills Foundation Innovator Grant

Proposal: "When High- Dimensional Data Goes Missing"

Total award estimate: \$50,000

07/2024-08/2025

ACHA Charles Chui Young Researcher Best Paper Award

Best Paper Award for "On the number of faces and radii of cells induced by Gaussian spherical tessellations", jointly awarded with Eric Lybrand

2022

Rising Stars in Computational and Data Science

Rising Stars University of Texas, Austin

04/2019

UC Chancellor's Postdoctoral Fellowship

2018-2019

Faculty Mentor: Rayan Saab, University of California, San Diego	
US Junior Oberwolfach Fellow	10/2018
Mathematisches Forschungsinstitut Oberwolfach	
Dissertation Fellowship	2017-2018
Claremont Graduate University	
Intellisis Fellowship	2017-2018
Computational Science Research Center at San Diego State University	
Intellisis Fellowship	2017
Computational Science Research Center at San Diego State University	
Los Alamos National Laboratory Poster Award	2017
Applied Computational Science and Engineering Student Conference (ACSESS)	
Computational Science Research Center at San Diego State University	
Outstanding Poster Presentation	2013
MAA Undergraduate Poster Session at the Joint Mathematics	
Poster: "Linking Social Media to Crime and Disorder", jointly awarded with Y. Hu	

Research Mentoring

Ph.D. Advising

Jenny Tran , Graduate Student, UC Irvine	09/2023-current
Emeric Battaglia , Graduate Student, UC Irvine	09/2023-current
Kathryn Dover, Ph.D. , Graduate Student, UC Irvine	09/2019-06/2022
• Thesis: Does t-SNE See Curves? Theory and Practice • First position: Lead Data Scientist at Syniverse • Co-chaired with Roman Vershynin	

Graduate Student Advising

Math 299 - Natanael Alpay , Graduate Student, UC Irvine	03/2025-06/2025
• Topic: Stochastic gradient descent and kernel regression	
Math 299 - Nigus Teklehaymanot , Graduate Student, UC Irvine	03/2025-06/2025
• Topic: Tensor-based singular value decomposition	
Math 299 - Ashwin Prabu , Graduate Student, UC Irvine	09/2024-03/2025
• Topic: Frontal Slice Descent methods for large-scale tensor systems	

Undergraduate Student Advising

Math 199 - Supervised Research Greedy Methods for Linear Feasibility	2024-2025
• James Nguyen (UC Irvine, 2025) • Adityakrishnan Radhakrishnan (UC Irvine, 2025) • Oleg Presnykov (UC Irvine, 2026)	
Math 199 - Supervised Research Oblique Iterative Projection Methods	2022-2024
• Emi Cervantes (UCI, 2024; First position: PhD in Statistics at UPenn) • Jenny Tran (UCI, 2024; First position: PhD in Mathematics at UCI) • Work presented at the Joint Mathematics Meeting 2024 (talk), Symposium for Women and Gender Minorities in Mathematics in Southern California 2024 (talk), Southern California Applied Mathematics Meeting 2024 (poster), and MAA Spring SoCalNev Sectional Meeting 2024 (talk)	
Math 199 - Supervised Research Coordinate-wise methods for block missing data	2022-2023

- Meha Patel (UCI, 2023; First position: M.S. in Statistics at CSU Long Beach)
- Sam Rath (UCI 2023; First position: M.S. in Math at San Diego State University)
- Chupeng Zheng (UCI, 2023; First position: M.S. in Data Science at U of Chicago)
- Work presented at Southern California Applied Math Symposium 2023 (poster) and MAA SoCalNev Sectional Meeting 2023 (talk)
- *Published article in SIAM Undergraduate Research Online (SIURO) "Mean Imputation and Stochastic Coordinate Descent for Linear Systems with Missing Data"*
doi:10.1137/23S1592018

Math 199 - Supervised Research Stochastic gradient methods for block missing data

2022-2023

- Chelsea Huynh (UCI Irvine, 2023; First position: M.S. in Computer Science at USC) and Michael Strand (UCI, 2023; First position: M.S. in Data Science at UC Irvine)
- Work presented at Southern California Applied Math Symposium 2023 (poster) and MAA SoCalNev Sectional Meeting 2023 (talk)
- *Published proceedings article in IEEE Allerton Conference on Communication, Control, and Computing "Block-missing data in linear systems: An unbiased Stochastic gradient descent approach for Linear Systems with Missing Data"*
doi:10.1109/Allerton58177.2023.10313429

IPAM - Research in Industry Projects (Academic Mentor) Traffic Shaping for Real Time Advertising with industry sponsor GumGum

Summer 2022

- Aresha Arshad (National University of Sciences and Technology, Pakistan, 2022)
- Gabriela Guerrero Galván (Pomona College, 2023)
- Nico Tripeny (Haverford College, 2023)
- Antonio Plascencia-Vargas (Grand Valley State University, 2023)

IPAM - Research in Industry Projects (Academic Mentor) Online Learning and Matching for Resource Allocation Problems with industry sponsor Alibaba

Summer 2019

- Andrea Boscovik (Amherst College, 2021)
- Qinyi Chen (UCLA, 2020)
- Dominik Kufel (University College London, 2021)
- Zijie Zhou (Purdue University, 2021)
- Work presented at Joint Mathematics Meetings 2020 (talk)
- *Published article in SIAM Undergraduate Research Online (SIURO) "Online Learning and Matching for Resource Allocation Problems"* doi:10.1137/19S1300534

UCLA - Research Experience for Undergraduate Students (Graduate Mentor) Topic Modeling and Lyme Disease Data

Summer 2018

UCLA - Research Experience for Undergraduate Students (Graduate Mentor) Matrix Completion and Topic Modeling for Lyme Disease Data

Summer 2017

Teaching

University of California, Irvine Instructor

2019-current

- Math 13 - Introduction to Abstract Mathematics (W23, SP22)
- Math 105A - Numerical Analysis I (F25, F23)
- Math 105AL - Numerical Analysis I Computing Lab (F25)
- Math 105B - Numerical Analysis II (W24)
- Math 130A - Probability and Stochastic Processes I (SP22, SP21, F20, W20, F19)
- Math 130B - Probability and Stochastic Processes II (W21)
- Math 140A - Introduction to Real Analysis (F21)
- Math 225B - Graduate Numerical Analysis (W25)
- Math 225C - Graduate Numerical Linear Algebra and Special Topics (SP25)
- AISCI 220 - Methods in Machine Learning and AI (FA26)

MSRI - Graduate Summer School Teaching Assistant	Summer 2018
• Representations in High-Dimensional Data	
San Diego State University, Department of Mathematics Teaching Assistant	2016-2018
• Math 151 - Calculus 2: Integration Techniques	
Claremont McKenna College, Department of Mathematics Graduate Tutor	2015-2017
• Math 252 - Statistical Theory, Math 30 - Calculus 1 Differentiation Techniques, CS 55 - Discrete Mathematics and Introduction to Probability	
• CCMS Coding lab: Matlab, R, C++, Latex	

Professional Activities

UC Irvine STEM Equity Learning Community	2023-2024
Participated in STEM learning community to discuss equity issues in lower-division STEM courses, particularly Math 13: Introduction to Abstract Mathematics as a part of the Sloan Equity and Inclusion in STEM Introductory Courses program. Presented findings and a call-to-action to the department in Spring 2024.	
Noticing in Mathematics for Student Success (NIMSS) Training	2022-2023
Participated in training provided by UC Irvine Department of Education, which focuses on the study and development of video-annotation curriculum to enhance Instructors' understanding of student thinking in proof-transition courses	
Online Remote Teaching Training	2020
Completed Division of Teaching Excellence & Innovation Online Remote Teaching Training.	

Talks and Presentations

Conferences

• Conference on Learning Theory, San Diego, CA <i>Presented accepted work "Efficient Tensor Completion with Nearly-Linear Sample Complexity"</i>	July 2026
• The 8th International Conference on Econometrics and Statistics, Tokyo, Japan <i>Session on Stochastic iterative methods for statistical learning</i>	August 2025
• The 26th Conf of the International Linear Algebra Society, Kaohsiung, Taiwan <i>Session on New methods in numerical multilinear algebra</i>	June 2025
• The 2nd Conf on Random Matrix Theory and Numerical Linear Algebra, Seattle, WA <i>Conference speaker</i>	June 2025
• SIAM Conference on Computational Science and Engineering, Fort Worth, TX <i>Applications of Tensor Methods for Computational and Data Sciences</i>	March 2025
• Practicum for Undergraduate Mathematicians, Los Angeles, CA <i>Workshop speaker</i>	November 2024
• MAA Mathfest, Indianapolis, IN <i>Iterative and Sketching Approaches for Linear Systems and Beyond</i>	August 2024
• SIAM Conference on Imaging Science, Atlanta, GA <i>Recent Challenges and Developments on Multi-way Data Analysis</i>	May 2024
• SIAM Conference on Applied Linear Algebra, Paris, France <i>Iterative Methods for Inverse Problems and Uncertainty Quantification</i>	May 2024
• AWM Research Symposium <i>Session on Tensor Methods for data modeling</i>	October 2023
• International Congress on Industrial and Applied Mathematics, Tokyo, Japan <i>Optimal and Efficient Algorithms for Inverse Problems</i>	August 2023
• SIAM Conference on Optimization, Seattle, WA <i>Randomized Methods for Linear Systems: Row-Action and Coordinate Descent</i>	June 2023

- Center for Approximation and Mathematical Data Analytics Conference, Texas A&M
Conference speaker May 2023
- AMS Spring Western Sectional Meeting, Fresno, CA
Special Session on Women in Mathematics May 2023
- Pacific Rim Mathematical Association Congress, Vancouver, Canada
Mathematics of Information December 2022
- SIAM Mathematics of Data Science, San Diego, CA
Tensor Methods and Applications to Real-World Data September 2022
- International Conference on Continuous Optimization, Lehigh University
Optimization for Data Science and Machine Learning July 2022
- Women in Inverse Problems Workshop, Virtual
Conference speaker December 2021
- SIAM Conference on Applied Linear Algebra, Virtual
Random Matrices and Numerical Linear Algebra May 2021
- SIAM Conference on Computational Science and Engineering, Fort Worth, TX
Computational Linear Algebra in Data Analysis March 2021
- ICERM Computational Stats and Data-Driven Models Workshop, Providence, CA
Conference speaker April 2020
- Women in Mathematics, Southern California, Irvine, CA
Conference speaker March 2020
- Joint Mathematics Meeting, Boulder, CO
Special Session on Mathematical and Computational Research in Data Science January 2020
- Joint Mathematics Meeting, Boulder, CO
Special Session on The Kaczmarz Algorithm with Applications January 2020
- Rising Stars in Computational and Data Science, Austin, TX
Conference speaker April 2019
- AWM Research Symposium, Houston, TX
Applied and Computational Harmonic Analysis April 2019
- AWM Research Symposium, Houston, TX
WISDM: Women in the Science of Data and Mathematics April 2019
- AMS Central and Western Spring Meeting, Honolulu, HI
Special Session on Sparsity, Randomness, and Optimization March 2019
- SIAM Conference on Computational Science and Engineering, Spokane, CA
AWM Workshop: Data Science and Mathematics February 2019
- Asilomar Conference on Signals, Systems, and Computers, Pacific Grove, CA
Conference speaker October 2018
- Women in Mathematics Southern California Symposium, Los Angeles, CA
Conference speaker February 2017

Seminars

- Applied Mathematics Seminar, Michigan State University April 2026
- RNLA Tutorial, Institute for Computational and Experimental Mathematics February 2026
- Probability Seminar, Northwestern University May 2025
- Applied Mathematics Seminar, University of California, San Diego April 2025
- Applied Mathematics Seminar, University of California, Santa Barbara February 2025
- Simple Words, Virtual Seminar November 2024
- Statistics and Data Science Seminar, Rice University October 2024
- Statistics and Data Science Seminar, Northwestern University April 2024
- Probability and Combinatorics Seminar, University of California, Irvine April 2024
- Mathematical Data Science, Purdue-University of Maryland-Penn State Jointly April 2024
- Probability Seminar, University of Southern California September 2023
- Data-oriented Math & Statistical Sciences Seminar, Arizona State University September 2023
- Graduate Seminar, University of California, Irvine May 2023
- Women and Math Young Researcher Seminar, Institute for Advanced Study May 2022
- CRM Applied Math Seminar, University of Montreal March 2022

• Young Data Science Research Seminar, ETH Zurich	February 2022
• One World MINDS Seminar, Virtual Seminar	September 2021
• CMSE Brown Bag Talks, Michigan State University	November 2020
• Graduate Seminar, CSU Channel Islands	May 2020
• Applied Mathematics Seminar, University of California, Merced	September 2019
• Undergraduate Colloquium, University of California, San Diego	March 2019
• Mathematical Sciences Applied Math Seminar, Claremont Colleges Consortium	February 2019
• Center for Computational Mathematics, San Diego State University	December 2018
• Mathematics of Data and Decisions at Davis, University of California, Davis	December 2018
• Summer Colloquium Series, San Diego State University	July 2018
• Level Sets Seminar, University of California, Los Angeles	May 2018
• GFA Young Researchers Seminar, Mathematical Sciences Research Institute	November 2017
• CS Theory Lunch, Stanford University	October 2017
• Algebra, Number Theory, and Combinatorics Seminar, Claremont Consortium	October 2017
• Graduate Seminar, Claremont Graduate University	April 2017

Colloquia

• Mathematical Sciences Mathematics Colloquium, Claremont Colleges Consortium	November 2025
• Department of Mathematics, University of Central Florida	April 2024
• Department of Mathematics, Wellesley College	December 2022
• Mathematics and Statistics Seminar, San Jose State University	September 2017

Posters

• AWM Research Symposium, Minneapolis, MN	June 2022
• Information Theory and Applications, La Jolla, CA	February 2018
• Data Institute Conference at University of San Francisco	October 2017
• Joint Mathematics Meeting, Baltimore, MD	January 2014
• Mathematical Sciences Session, Claremont Consortium, Claremont, CA	September 2014
• AFSOR MURI Fall Meeting, Los Angeles, CA	November 2014
• Asilomar Conference on Signals, Systems, and Computers, Pacific Grove, CA	November 2014
• Optimization in ML Workshop, NeurIPS, Montreal, Canada	December 2015
• February Fourier Talks, University of Maryland, College Park	February 2016
• ACCESS Meeting, Computational Science Research Center, San Diego, CA	April 2016
• Southern California Applied Math Symposium, Claremont, CA	June 2016
• ACCESS Meeting, Computational Science Research Center, San Diego, CA	April 2017
• Gene Golub SIAM Summer School, Germany	June 2017
• Asilomar Conference on Signals, Systems, and Computers, Monterey, CA	November 2019

Outreach and other presentations

• MathCounts Orange County Chapter Competition, Irvine, CA, <i>Faculty Lecturer</i>	February 2025
• UCI Math Circle, Faculty lead	February 2025
• Undergraduate Math Club - Data Science in Math, University of California, Irvine	November 2023
• Computational Science Research Center, San Diego State University, Alumni Panel	April 2023
• UCI Math Circle, Faculty lead	April 2024
• Research in Industrial Projects (RIPS) Celebration 2023, IPAM, Alumni Panel	August 2023
• Math, Computational and Systems Biology Recruitment, UC Irvine, Career Panel	February 2023
• Department of Mathematics Postdocs, UC Irvine, Career Panel	November 2022
• Computational Science Research Center, San Diego State University, Career	February 2019
• Representations of High Dimensional Data, MSRI, Career Panel	July 2018
• The Mathematics of Data Science, MSRI, Graduate Student Panel	July 2018

Service

Conferences and Workshops

- AIMS Workshop “Fairness and Foundations in Machine Learning” July 2026
- ICERM Workshop, “Randomized Numerical Linear Algebra (RNLA)” February 2026
- IPAM Research Collaborations, “Randomized Numerical Linear Algebra (RNLA)” August 2025
- The 26th Conference of the International Linear Algebra Society, Minisymposium June 2025
- UC Irvine Mathematics of Data, Dynamics, and Life Sciences March 2025
- Joint Mathematics Meetings, Special Session January 2025
- SIAM Mathematics of Data Science, Minisymposium October 2024
- SIAM Linear Algebra and Applications, Minisymposium May 2024
- Sampling Theory and Applications, Minisymposium July 2023
- Southern California Applied Math Research Symposium (SoCAMS) April 2023
- Joint Mathematics Meeting, Special Session January 2022
- Optimal Transport: Theory, Computation, and Applications Workshop June 2022
- Southern California Applied Math Research Symposium (SoCAMS) May 2022
- Women in Math, Southern California (WiMSoCal) March 2021
- Information Theory and Applications, Minisymposium February 2020
- AWM UCI Chapter Organizer (Postdoctoral) 2020-2021
- SIAM Computational Science and Engineering Conference, Minisymposium February 2019
- University of San Francisco, Data Science Conference, Special Session February 2019
- Joint Mathematics Meeting, Special Session January 2019
- CGU Graduate Student Seminar 2016-2018

Press

- Interviewed for Her Math Story, a platform for showcasing the careers and journeys of modern-day mathematicians, <https://hermathsstory.eu/anna-ma/> January 2025
- Interviewed for article “Greed Works: An Improved Analysis of Sampling Kaczmarz–Motzkin” featured in SIAM Youtube channel, https://www.youtube.com/watch?v=L0zJf_pdpZ0 June 2024
- Research featured in Claremont Graduate University Alumni Spotlight, <https://www.cgu.edu/news/2023/05/anna-ma-phd-in-computational-science-2018-streamlines-the-algorithm/> May 2023
- Co-organizer for SoCAMS conference featured in SIAM News Blog: “Mathematicians Convene at 2023 Southern California Applied Mathematics Symposium”, <https://www.siam.org/publications/siam-news/articles/mathematicians-convene-at-2023-southern-california-applied-mathematics-symposium/> May 2023

Other Activities

- **Coffee/Lunch with a Math Professor Organizer** This program facilitates students interacting more personally with a professor over lunch or coffee. It encourages students and faculty to meet more outside the usual office hours and classroom settings. This program has been an enormous success, reaching over 200 undergraduate students since its inception. 2021-current
- **Cal-Bridge Mentor** The Cal-Bridge program aims to broaden participation by Californians in STEM PhD Programs. As a Cal-bridge mentor, I worked one-on-one with an undergraduate student at a California State University to bridge their time between Cal State and UC and prepare them for applying to PhD Programs. 2024-current
- **Girls’ Angle Bulletin Contributing Writer** Girls’ Angle is a nonprofit math club founded in 2007. The goal of Girls’ Angle is to foster and nurture girls’ interest in mathematics and empower them to tackle any field, no matter the level of mathematical sophistication. I was a contributing writer for the bimonthly magazine Girls’ Angle Bulletin. 2021-2023

Reviewing and editing

- **Proposals:** NSF, UC Postdoctoral Presidential Fellowship, Simons Foundation
- **Editing:** Co-editor of topical collection on tensor methods in *La Matematica*, 2024
- **Manuscripts and Conferences:** AAAI 2025 Workshop ColorAI, *Advances in Computational Mathematics*, *Applied and Computational Harmonic Analysis*, *Applied Mathematics Letters*, *Applied Numerical Mathematics*, *BIT Numerical Mathematics*, *Calcolo*, *Electronic Transactions on Numerical Analysis*, *IEEE Letters in Signal Processing*, *IEEE Transactions on Signal Processing*, *Inverse Problems and Imaging*, *Journal of Computational and Applied Mathematics*, *Journal of Machine Learning Research*, *Journal on Scientific Computing*, *Linear and Multilinear Algebra*, *Machine Learning*, *Numerical Algorithms*, *Numerical Linear Algebra*, *Numerical Linear Algebra and Applications*, *Numerical Mathematics: Theory, Methods and Applications*, *Numerische Mathematik*, *Proceedings of the National Academy of Sciences*, *Sampling Theory and Applications (SampTA)*, *SIAM Journal on Data Science*, *SIAM Journal on Imaging Sciences*, *SIAM Mathematics of Data Sciences*, *SIAM Matrix Analysis and Applications*, *Signal Processing*, *Signal Processing Letters*, *Signal Processing with Adaptive Sparse Structured Representations (SPARS)*